INCH-POUND

MS27222N
22 October 2020
SUPERSEDING
MS27222M
27 November 2003

PERFORMANCE SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, 50 AMPERES, 3 PST, N. O. TYPE I, HERMETICALLY SEALED

INACTIVE FOR NEW DESIGN AFTER 15 NOVEMBER 2002.
NO SUPERSEDING SPECIFICATION.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-PRF-6106.

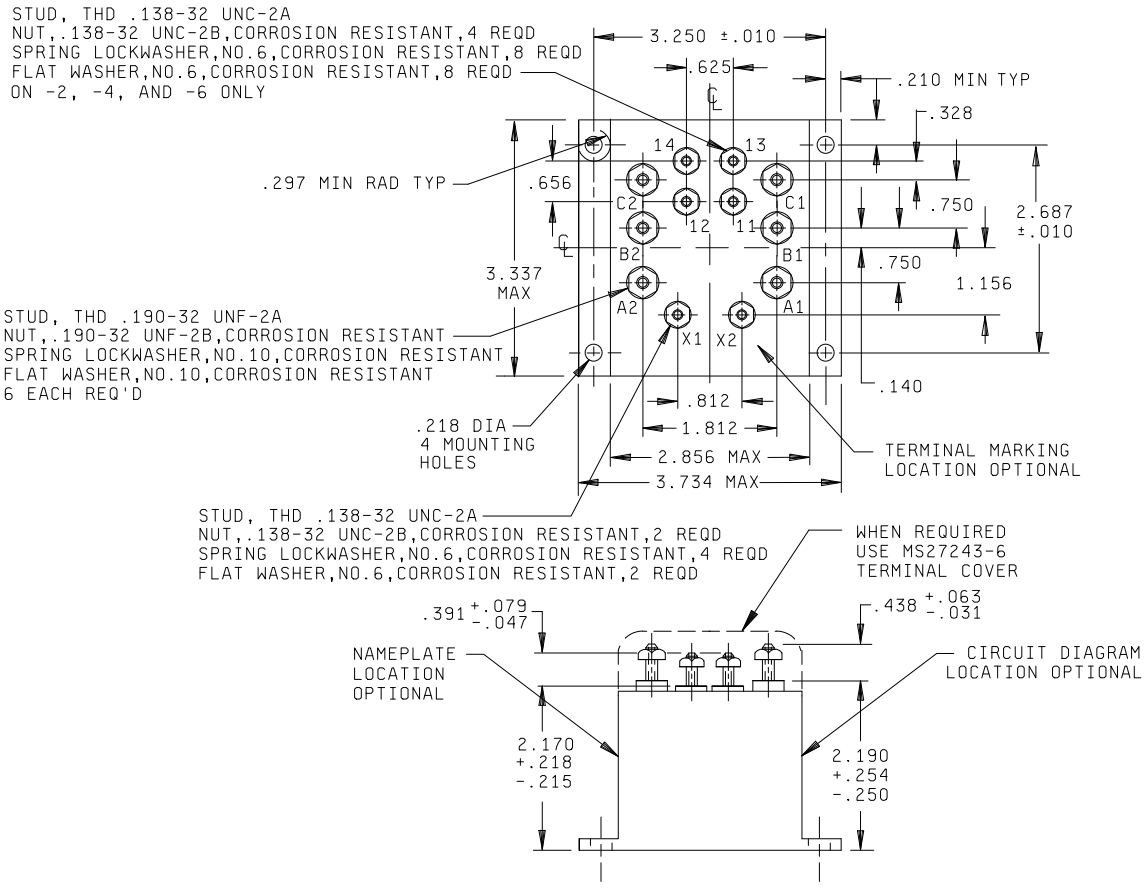
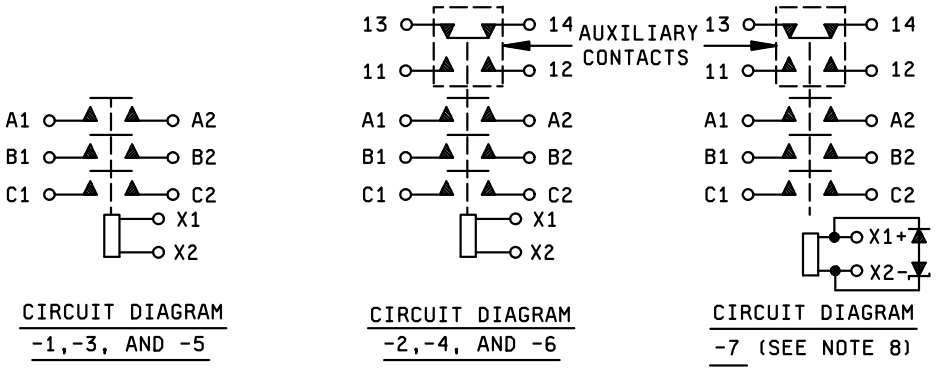


FIGURE 1. Dimensions and configuration.

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Inches	mm	Inches	mm	Inches	mm
.010	0.25	.218	5.54	.812	20.62
.031	0.79	.250	6.35	1.156	29.36
.047	1.19	.254	6.45	1.812	46.02
.063	1.60	.297	7.54	2.170	55.12
.079	2.01	.328	8.33	2.190	55.63
.138	3.51	.391	9.93	2.687	68.25
.140	3.56	.438	11.13	3.250	82.55
.190	4.83	.625	15.88	3.337	84.76
.210	5.33	.656	16.66	3.734	94.84
.215	5.46	.750	19.05		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.031 (0.79 mm).
4. Coil and auxiliary terminals may use additional washer for terminal seat.
5. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.
6. Referenced Government documents of the issue listed in that issue of the Acquisition Streamlining and Standardization Information System (ASSIST) specified in the solicitation forms a part of this standard to the extent specified herein.
7. Terminal strength: High temperature pull and torque test not applicable.
8. When semiconductors are required, JANTX or equivalent screened semiconductors shall be used. Relays using suppression devices shall continue to operate should the suppression circuit be in a failure mode. Diodes shall have a peak inverse voltage of 600 V dc minimum.
9. For detail information, see tables I through V.

FIGURE 1. Dimensions and configurations – Continued.

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REQUIREMENTS:

Dimensions, configuration, and circuit diagrams: See figure 1.

Dash numbers and general characteristics: See table I.

Contact data:

Load ratings: See table II.

Maximum contact drop:

Initial: 0.150 volts.

After life test: 0.175 volts.

Overload current (NO): 400 amperes.

Rupture current (NO): 500 amperes (only 5 operations rupture after minimum current test).

Coil data: See table III.

Duty rating: Continuous.

RFI specification: MIL-STD-461 (applicable to coil circuits of ac operated relays).

Electrical data:

Minimum insulation resistance:

Initial: 100 megohms.

After life or environmental test: 50 megohms.

Dielectric strength (sea level): 2 - 5 seconds (see table IV).

Dielectric strength (altitude): 1 minute (see table V).

Environmental characteristics:

Temperature range: -70°C to +125°C.

Maximum altitude rating: 80,000 feet.

Shock g level: 59 g's:

Duration: 6 - 9 ms.

Maximum duration contact opening: 2 ms.

Vibration (sinusoidal): See table VI.

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Non-operate:

G level: 15 g's.

Frequency range curve: 20 to 2,000 Hz.

Acceleration: 15 g's.

Conformance inspection.

Performance of groups B and C tests are not applicable.

Qualification by similarity: See MIL-PRF-6106.

Part or Identifying Number (PIN): MS27222- (plus dash number from table I).

TABLE I. Dash numbers and characteristics.

Dash number MS27222-	Type	Coil type	Terminal type	Mounting or mating socket	Auxiliary terminals	Max weight wt. lbs. 1/
1	I	dc	Lug	N/A	None	1.30
2	I	dc	Lug	N/A	With	1.35
3	I	ac	Lug	N/A	None	1.30
4	I	ac	Lug	N/A	With	1.35
5	I	ac	Lug	N/A	None	1.30
6	I	ac	Lug	N/A	With	1.35
7	I	dc	Lug	N/A	With	1.30

1/ Weights include terminal barriers and covers.

TABLE II. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/				See appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	50	50	---	5	5	50	---	5	5	50	---	---	---	
Inductive	10	50	---	5	5	50	---	5	5	50	---	---	---	
Inductive														
Motor	50	50	---	---	---	50	---	---	---	50	---	---	---	
Lamp	---	---	---	1	1	---	---	1	1	---	---	---	---	
Transfer load														2/
Mechanical life reduced current	100	12.5	---	1.25	1.25	12.5	---	1.25	1.25	12.5	---	---	---	
Mixed loads	50	Applicable in accordance with MIL-PRF-6106												

1/ Absence of value indicates relay is not rated for 3-phase application.
2/ Transfer load indicates relay is suitable for transfer between unsynchronized ac power supplies at rating indicated.

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TABLE III. Operating characteristics.

PIN MS 27222-	Coil data												Time - (milliseconds maximum)				
	Coil	Nominal			Max		Max pick-up voltage			Hold voltage 2/	Drop out voltage 2/	Operate 3/	Release 4/	Contact Bounce			
														Main		Aux	
		Volts 1/	Freq Hz	Res Ω ±10%	Volts	Amp	Normal 2/	High temp test	Cont current test					NO	NC	NO	NC
1	X1,X2	28	dc	51	29	.6	18	21	22.5	7.0	1.5	25	10	2	---	---	---
2	X1,X2	28	dc	51	29	.6	18	21	22.5	7.0	1.5	25	10	2	---	4	4
3	X1,X2	115	60/ 400	N/A	124	.2	90	100	104	40	10	25	50	2	---	---	---
4	X1,X2	115	60/ 400	N/A	124	.2	90	100	104	40	10	25	50	2	---	4	4
5	X1,X2	115	400	N/A	124	.2	90	100	104	40	10	25	50	2	---	---	---
6	X1,X2	115	400	N/A	124	.2	90	100	104	40	10	25	50	2	---	4	4
7 5/	X1,X2	28	dc	51	29	.6	18	21	22.5	7.0	1.5	25	10	2	---	4	4

1/ CAUTION: Use of any coil voltage less than rated coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With rated coil voltage.
4/ From rated coil voltage.
5/ Coil suppression (transient voltage back EMF) 43 V dc maximum.

Table IV. Dielectric strength (sea level).

Dielectric strength	28 V dc		115 V ac	
	Initial	After life tests	Initial	After life tests
Coil to case	1,250	1,000	1,500	1,125
Aux contacts	1,250	1,000	1,500	1,125
All other points	1,250	1,000	1,800	1,350

TABLE V. Dielectric strength (altitude).

Dielectric strength	28 V dc	115 V ac
Coil to case	500	500
Aux contacts	500	500
All other points	700	700

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TABLE VI. Vibration level (sinusoidal).

PIN MS27222-	5 - 10 Hz	10 - 55 Hz	55 - 750 Hz	750 - 1500 Hz
1, 3, and 5	.06 DA	.06 DA	15 g's	12 g's
2, 4, and 6	.06 DA	.06 DA	10 g's	10 g's

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. In addition to MIL-PRF-6106, this document references the following:
MS27243J MIL-STD-461

Custodians:
Navy - AS
Air Force - 85
DLA - CC

Review activity:
Navy - EC

Preparing activity:
DLA - CC

(Project 5945-2020-061)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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